Examiner only

2. (a) When lithium reacts with water in a large beaker hydrogen gas is released.

Lithium hydroxide solution is also formed. This turns universal indicator purple.

(i) Tick (✓) the box next to the description of what is seen when lithium reacts with water in a large beaker. [1]

lithium melts into a ball and sinks ☐

lithium fizzes and moves around the surface of the water ☐

lithium catches fire and burns with a blue flame ☐

(ii) Tick (✓) the box that describes lithium hydroxide solution. [1]

neutral ☐ acid ☐ alkali ☐

(iii) Lithium hydroxide contains Li^+ and OH^- ions.

Circle the correct formula for lithium hydroxide. [1]

liOH **LiOH** **$Li(OH)_2$** **Li_2OH**

Examiner only

(b) Lithium reacts with chlorine to form lithium chloride.

(i) Tick (✓) the box next to the correct balanced equation for the reaction. [1]

$2Li + Cl_2 \longrightarrow 2LiCl_2$ ☐

$2Li + Cl_2 \longrightarrow 2Li_2Cl$ ☐

$2Li + Cl_2 \longrightarrow 2LiCl$ ☐

(ii) Anwen was asked to use a flame test and a silver nitrate test to identify lithium chloride.

Circle the expected observation for each test. [2]

Flame test	**Silver nitrate test**
green flame	yellow precipitate
red flame	blue precipitate
lilac flame	white precipitate

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(c) Lithium reacts with oxygen to form lithium oxide.

Tick (✓) the box next to the calculation used to find the relative formula mass (M_r) of lithium oxide, Li_2O. [1]

$A_r(Li) = 7$ $A_r(O) = 16$

7 + 7 + 16 ☐

7 + 16 ☐

7 + 7 + 16 + 16 ☐

7 + 16 + 16 ☐

7